Objective 1

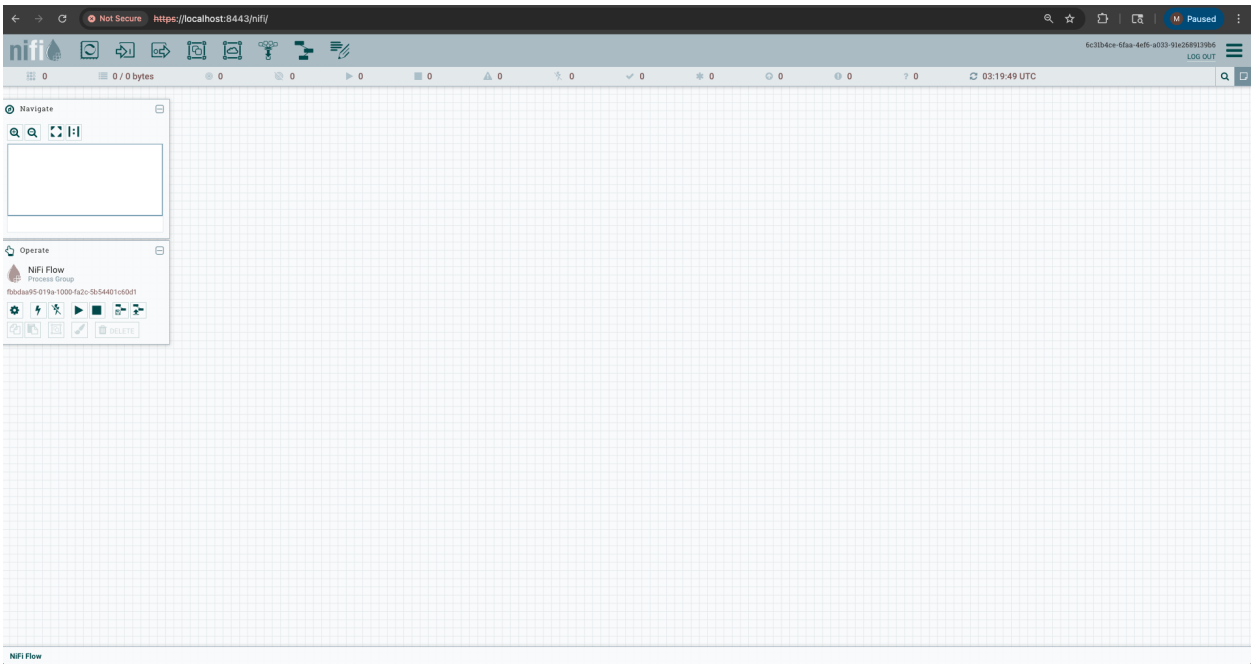
Apache NiFi is a data integration and automation platform designed to move and manage data between systems in a reliable and controlled way. The fundamentals video explains that NiFi follows a flow-based programming model, where data moves through a series of connected processors on a user interface. Each piece of data traveling through the system is called a FlowFile. A FlowFile contains both the actual data content and metadata attributes that describe it. Processors act on FlowFiles to perform tasks such as generating data, transforming formats, filtering records, routing based on conditions, or sending data to external systems. This visual approach makes complex data pipelines easier to design, monitor, and troubleshoot.

NiFi tracks every FlowFile as it moves through the system, providing detailed insight into where the data came from, how it was transformed, and where it was sent. This transparency is essential for debugging, auditing, and ensuring data governance standards are met. The ability to monitor and trace data movement makes NiFi especially valuable in environments where compliance and accountability are important.

These foundational concepts directly connect to this assignment because we are building data flows in NiFi and integrating them with Solr. NiFi is responsible for generating and processing log data, while Solr serves as the indexing and search engine that stores and queries that data. By sending processed logs from NiFi into a Solr collection, we create a complete data pipeline that automates ingestion and enables powerful search capabilities. This demonstrates how NiFi and Solr work together to manage, process, and analyze data efficiently in a real-world system.

Objective 2

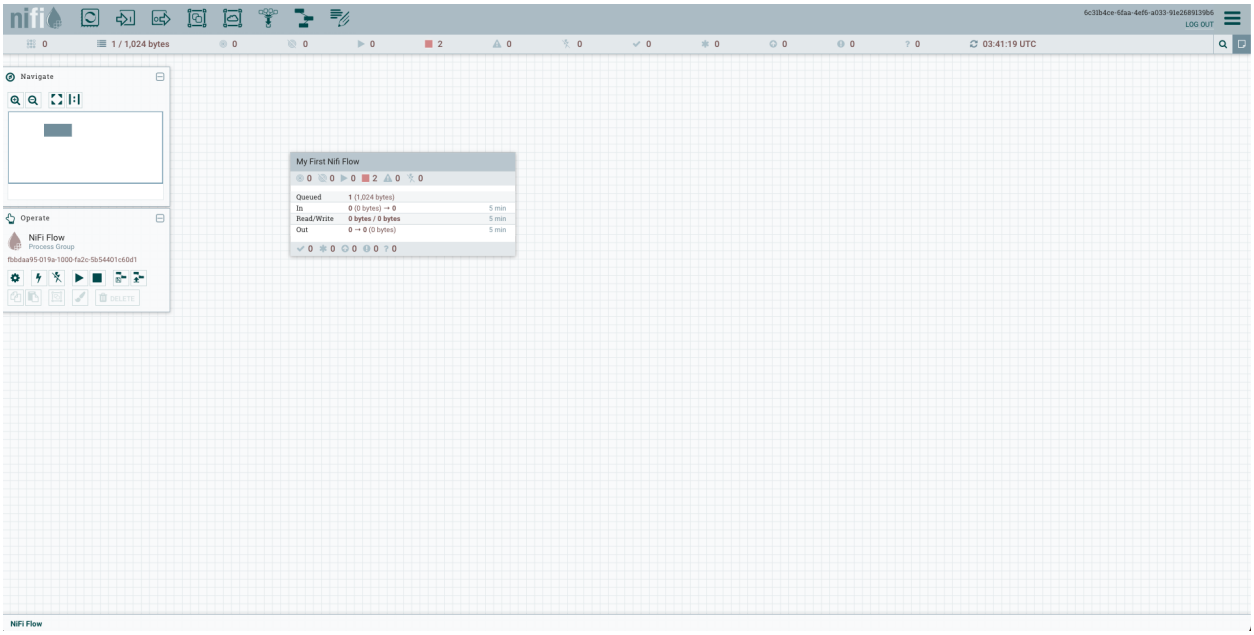
Deliverable 1



I started NiFi using the provided startup script and accessed the NiFi web interface through the browser. The UI confirms that NiFi is running because the canvas and controller options are visible.

Objective 3

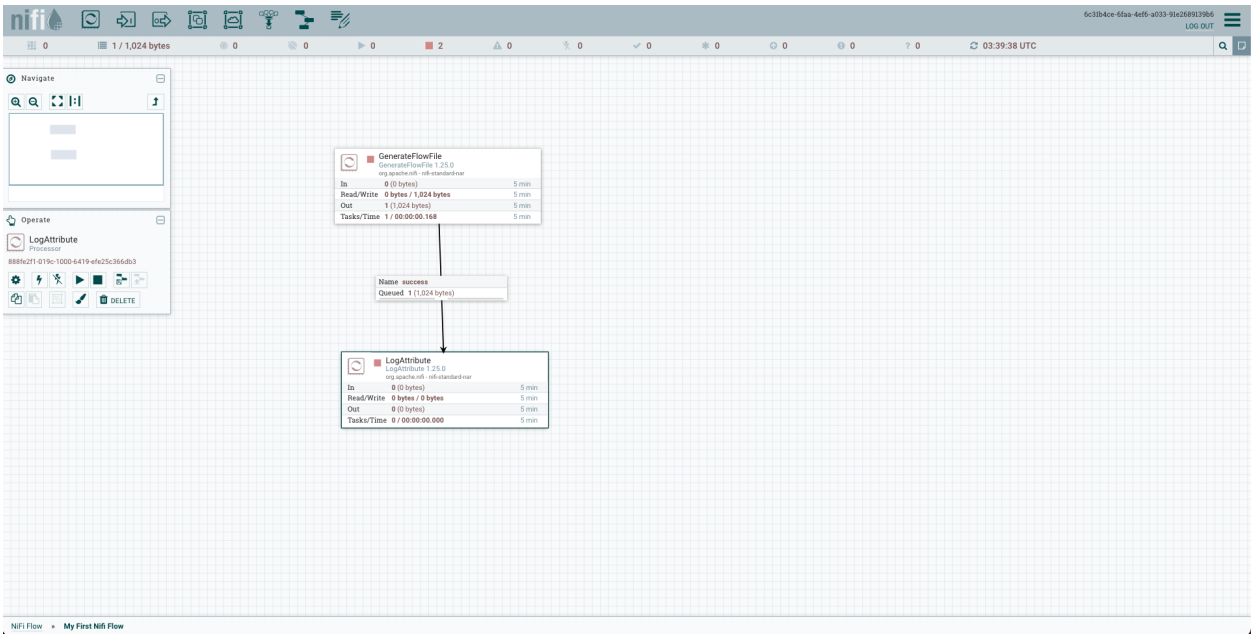
Deliverable 1



I created a parameter named My First Nifi Flow to store the Solr collection name. Parameter contexts allow centralized configuration, making flows easier to manage and update.

Objective 4

Deliverable 1



This flow generates a FlowFile and logs its attributes. The output confirms that NiFi is successfully creating and processing data through connected processors.

Objective 5

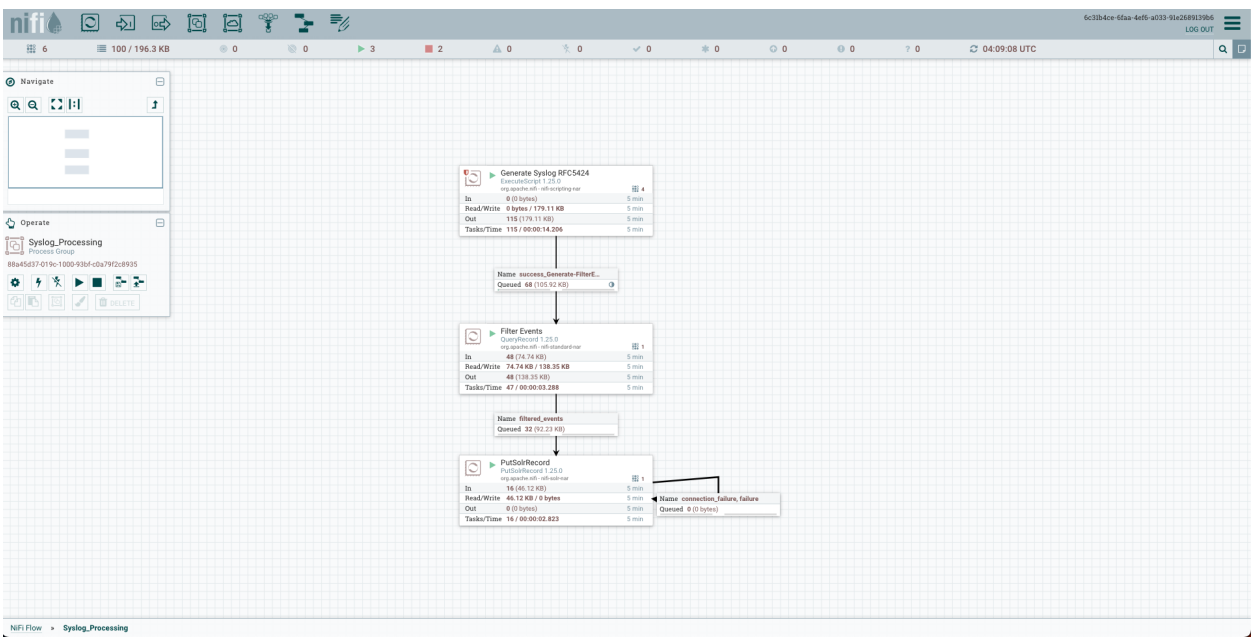
Deliverable 1

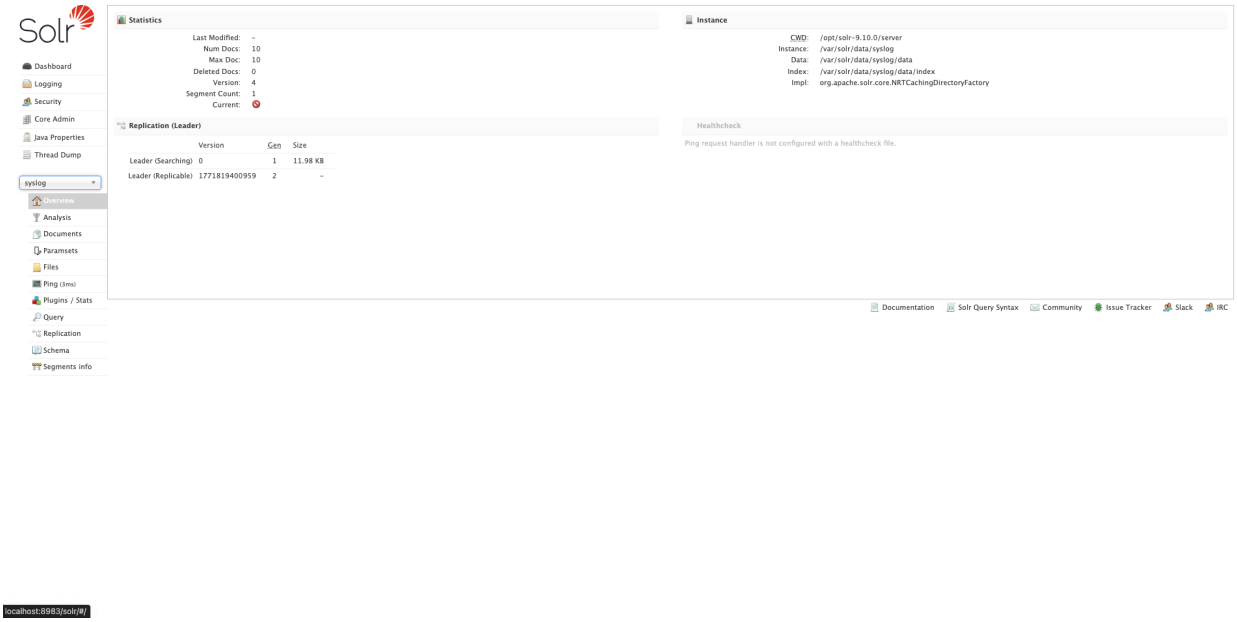
```
mahadfarah@dsc650-bigdata-mf:~/dsc650-infra/bellevue-bigdata/solr$ docker-compose up -d
solr_solr_1 is up-to-date
mahadfarah@dsc650-bigdata-mf:~/dsc650-infra/bellevue-bigdata/solr$ docker exec -it solr-solr-1 bash
Error response from daemon: No such container: solr-solr-1
mahadfarah@dsc650-bigdata-mf:~/dsc650-infra/bellevue-bigdata/solr$ docker exec -it solr-solr-1 bash
Error response from daemon: No such container: solr-solr-1
mahadfarah@dsc650-bigdata-mf:~/dsc650-infra/bellevue-bigdata/solr$ docker exec -it solr_solr_1 bash
solr@052f09249001:/opt/solr-9.10.0$ /opt/solr/bin/solr create -c syslog
Neither --zk-host or --solr-url parameters provided so assuming solr url is http://localhost:8983.
WARNING: Using _default configset. Data driven schema functionality is enabled by default, which is
         NOT RECOMMENDED for production use.

         To turn it off:
            curl http://localhost:8983/solr/syslog/config -d '{"set-user-property": {"update.autoCreateFields":"false"}}'
         Or:
            bin/solr config -c syslog -s http://localhost:8983 --action set-user-property --property update.autoCreateFiel
ds --value false

Created new core 'syslog'
solr@052f09249001:/opt/solr-9.10.0$
```

The syslog collection will store log data sent from NiFi, allowing it to be indexed and queried using Solr's search capabilities.

Objective 6

Deliverable 1

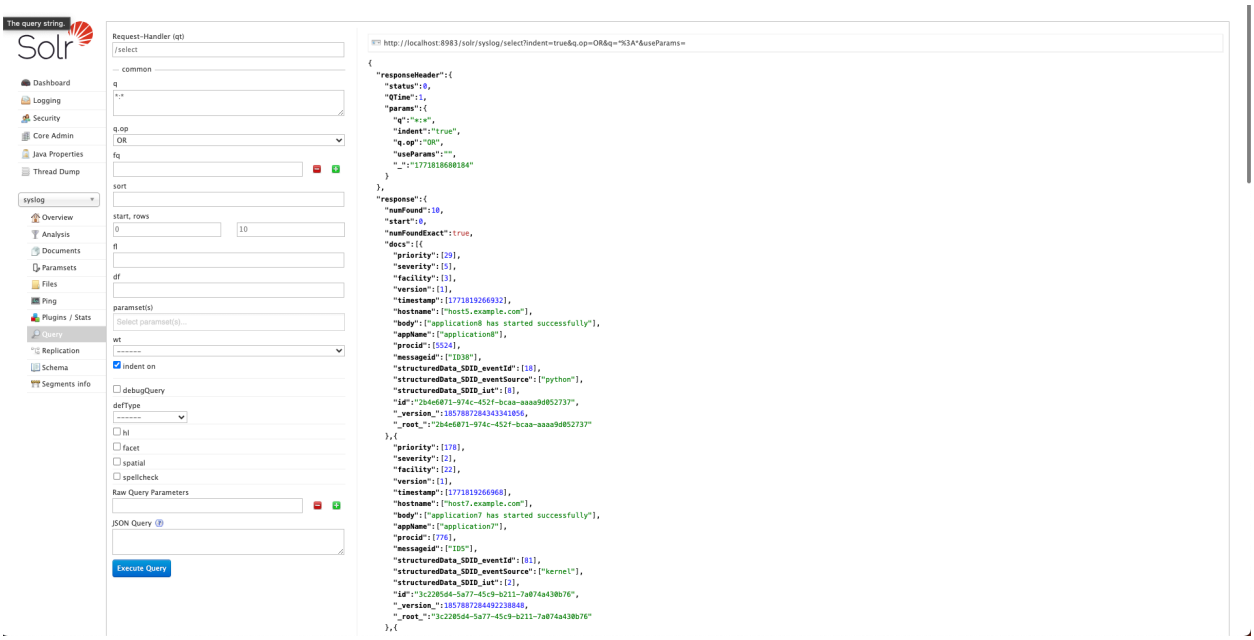




This flow generates and processes log data in NiFi and indexes it into Solr. The results confirm successful integration and data ingestion.

Objective 7

Deliverable 1



I queried the syslog collection using a match-all query to view indexed log entries. The results show that log data was successfully stored and is searchable in Solr.